Preliminary Project Proposal

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Abstract

This project proposes a novel Reinforcement Learning (RL)-based Brain-Computer Interface (BCI) framework for the intuitive and adaptive control of a robotic arm. Traditional BCI systems often rely on supervised learning classifiers with deterministic command mapping, which lack real-time adaptability to a user's changing intentions. To address this limitation, this system will employ a Deep Q-Network (DQN) or a more advanced RL algorithm to learn an optimal control policy directly from Motor Imagery (MI) Electroencephalography (EEG) signals. The project will leverage public EEG datasets (e.g., BCI Competition IV-2a) and a low-cost robotic arm as the physical platform. The core innovation lies in replacing the static classifier with an adaptive RL agent that optimises its control behaviour through trial-and-error interaction with the environment, thereby enhancing system responsiveness and accuracy. The system will be trained and evaluated primarily in a simulated environment, focusing on metrics such as task success rate and learning efficiency, and benchmarked against a baseline method using a deep learning classifier.

1 Introduction

1.1 Background and Motivation

Brain-Computer Interfaces (BCIs) provide a direct neural pathway for users to interact with external devices, such as robotic arms, showing immense potential in assistive technology and neurorehabilitation [1]. Among various modalities, non-invasive BCIs based on Motor Imagery (MI) from Electroencephalography (EEG) are widely studied due to their safety and convenience. However, the inherent low signal-to-noise ratio (SNR) and high inter-subject variability of MI-EEG signals pose significant decoding challenges [1].

Current state-of-the-art BCI systems, such as those based on deep learning models like CTNet, have achieved excellent offline accuracy in MI classification tasks [2]. These systems, however, typically operate on a "classify-then-map" paradigm: a fixed classifier decodes the EEG signal into a discrete intent, which is then mapped to a robot command via a deterministic rule [2]. While effective, this approach is fundamentally static and struggles to adapt to shifts in user state or dynamic environmental changes, limiting the intuitiveness and fluidity of control.

Reinforcement Learning (RL), which enables an agent to learn optimal strategies through trial-and-error interaction, offers a new paradigm for developing more adaptive and intelligent control systems. Recent studies have successfully applied RL to EEG signal processing for tasks such as classification [3] and feature selection [4], demonstrating the viability of RL frameworks for interpreting EEG signals. The core motivation of this project, therefore, is to advance the application of RL from these auxiliary roles to the direct task of robotic **control**. The goal is to develop a BCI system capable of adapting to user intent in real-time and learning an intuitive control policy, bridging the gap between current static BCI models and the demands of complex, real-world applications.

1.2 Aims and Objectives

The primary aim of this project is: **To design, implement, and evaluate a Reinforcement Learning-based BCI system that learns to directly and adaptively control a low-cost robotic arm using public MI-EEG datasets.**

To achieve this aim, the following objectives are set:

1. **Literature Review and System Setup:** Conduct a thorough review of RL algorithms for robotic control and BCI systems. Select a suitable public EEG dataset (e.g., BCI Competition IV-2a [2] or the GigaScience dataset [5]) and establish both physical and simulated environments for the robotic arm.

2. **Data Preprocessing and State Representation:** Develop a robust EEG signal preprocessing pipeline, including filtering and artifact removal. Design an informative state representation for the RL agent, drawing inspiration from related works [3].

3. **RL Framework Design:** Formalize the robotic control task as a Markov Decision Process (MDP). Define the agent's action space (e.g., discrete movement commands for the arm) and a reward function to guide the agent towards completing a specified task (e.g., reaching a target).

4. **Model Implementation and Training:** Implement a deep RL agent (e.g., DQN) with a policy network architecture suitable for spatio-temporal data, such as a CNN-LSTM hybrid [3]. Train the agent within the simulated environment.

5. **System Evaluation and Baseline Comparison:** Quantitatively evaluate the trained RL policy in the simulated environment on metrics including task success rate, learning efficiency, and convergence speed. Compare these results against a baseline system built on a supervised learning classifier [2] to demonstrate the advantages of the RL approach.

2 Project Methodology

1. **Literature Review:** The review will focus on deep reinforcement learning algorithms for robotics (DQN, PPO, SAC, etc.) and the application of RL in the BCI domain, with particular attention to [3] and [4].

2. **Hardware and Dataset Preparation:** A low-cost robotic arm, as recommended by the supervisor, will be used. A corresponding simulation environment will be created in PyBullet or Gazebo. The BCI Competition IV-2a dataset will be the primary data source, and a standard preprocessing pipeline involving band-pass filtering and ICA for artifact removal will be applied [5].

Figure 1: Figure:SO-ARM101

3. **RL Environment Construction:** A custom simulation environment conforming to the OpenAI Gym interface will be developed. In this environment, the state will be derived from preprocessed EEG trials (or their features), actions will map to discrete end-effector movements (e.g., up, down, left, right), and the reward function will be based on the distance between the end-effector and a target location.

4. **RL Agent Implementation:** A DQN agent will be implemented. The Q-network will feature a 1D-CNN combined with an LSTM, an architecture proven effective for extracting spatio-temporal features from EEG signals in RLEEGNet [3].

5. **Experimental Evaluation:**

   - **Training:** The DQN agent will be trained in the simulated environment to control the robotic arm to reach randomly generated targets, using EEG signals from the dataset as input.
   - **Baseline Comparison:** A baseline system will be implemented using a supervised classifier (e.g., EEGNet [4]) to classify EEG signals, with the output mapped to robot actions via a deterministic rule, mimicking the approach in [2].
   - **Evaluation:** The RL agent and the baseline system will be compared on identical test tasks based on success rate, average cumulative reward, and convergence speed.

3 Risk Management

- **Algorithmic Risk: RL Training Instability**
  - *Description:* The RL agent may fail to learn an effective policy due to the noisy nature of EEG signals and potentially sparse rewards.
  - *Likelihood/Impact:* High/High.
  - *Mitigation Strategy:* a) Implement reward shaping to provide denser feedback; b) Systematically tune hyperparameters using methods like the evolutionary search mentioned in [4]; c) Optimize the state representation using advanced feature extraction techniques such as the OVR-CSP method proposed in [3].

- **Data Risk: Dataset Quality and Applicability**
  - *Description:* Low SNR in public datasets or a mismatch between the task paradigm and the continuous control goal could hinder learning.
  - *Likelihood/Impact:* Medium/High.
  - *Mitigation Strategy:* a) Implement thorough signal preprocessing and artifact removal (e.g., ICA) [5]; b) Apply data augmentation techniques like the ”Segmentation-and-Recombination” method [2]; c) Compare the suitability of different datasets (e.g., BCI Comp IV-2a vs. GigaScience [5]).

- **Software/Hardware Risk: Simulation-to-Reality Gap**
  - *Description:* The policy trained in simulation may underperform when transferred to the physical robotic arm.
  - *Likelihood/Impact:* Medium/Medium.
  - *Mitigation Strategy:* a) Ensure the simulation environment accurately models the real robot’s dynamics; b) If time permits, attempt to fine-tune the policy on the physical hardware; c) Implement safety constraints (e.g., workspace and joint limits) as described in [2] to ensure safe operation on the physical hardware.

References

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[5] J.-H. Jeong, J.-H. Cho, et al., "Multimodal signal dataset for 11 intuitive movement tasks from single upper extremity during multiple recording sessions," *GigaScience*, vol. 9, no. 10, 2020.

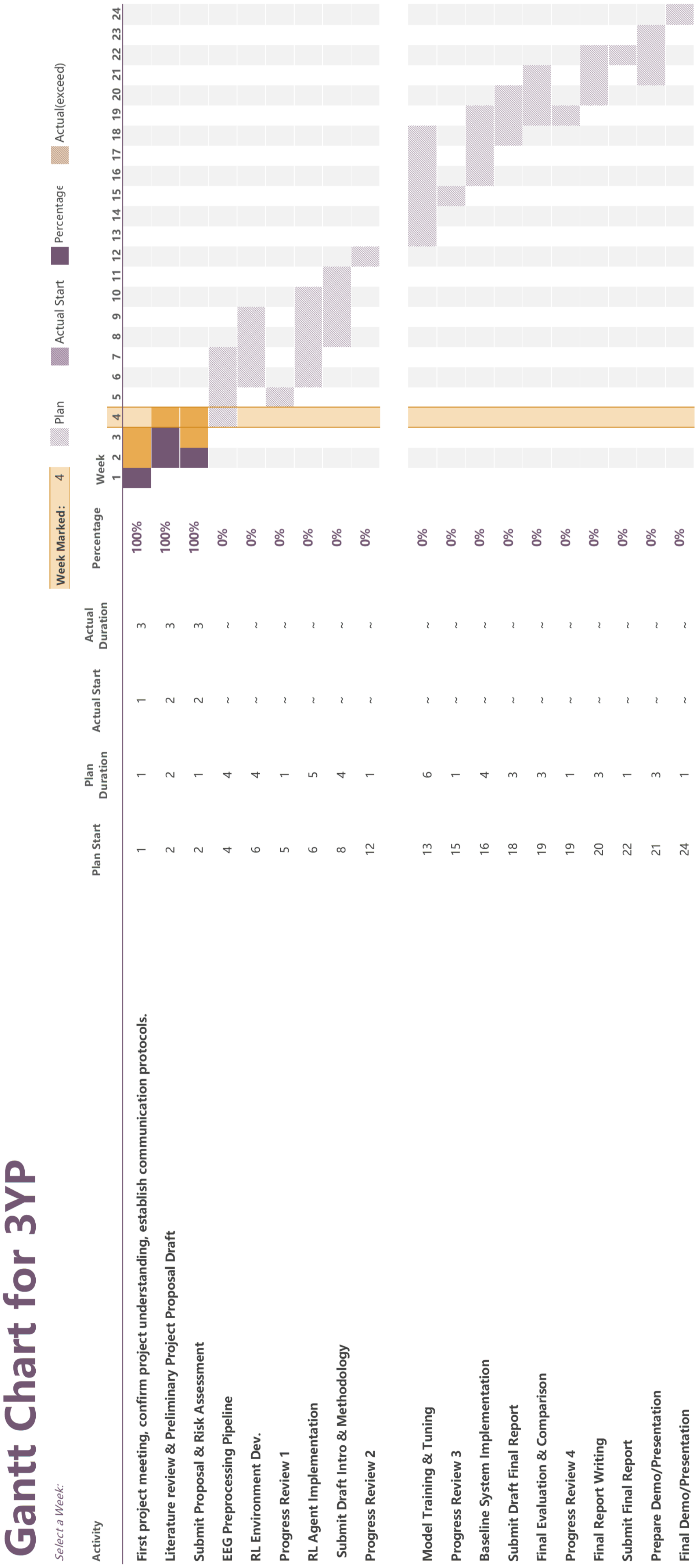

Figure 2: Figure:Gantt Chart